
TITANIC SURVIVAL PREDICTION

Machine Learning Classification using Python
Logistic Regression and Decision Tree Implementation

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1. Introduction

1.1 Introduction

Machine learning is widely used to identify patterns in data and make predictions about unseen observations. Classification is a supervised learning task in which a model learns from labelled examples and predicts a category for new inputs. The Titanic passenger dataset provides a practical example of binary classification, where the target variable indicates whether a passenger survived.

The proposed Titanic Survival Prediction System uses passenger information such as passenger class, sex, age, family-related attributes, fare, and embarkation port. The system preprocesses the data, trains classification models, evaluates their performance, and provides an interactive prediction interface using Streamlit.

The project implements Logistic Regression and Decision Tree classifiers. Logistic Regression provides a probabilistic classification baseline, while the Decision Tree learns a sequence of feature-based decision rules.

1.2 Problem Statement

The Titanic dataset contains passenger records with demographic, ticket, and travel-related attributes. The objective is to develop an automated machine learning system that can learn from historical passenger records and predict whether a passenger belongs to the survived or did-not-survive class.

The system should handle missing values, categorical attributes, and numerical attributes before training the models. It should also provide meaningful evaluation metrics so that the performance of the classifiers can be understood beyond accuracy alone.

1.3 Objectives

To analyze the Titanic passenger dataset using Python.

To identify and handle missing values in the dataset.

To perform exploratory data analysis of passenger survival.

To preprocess numerical and categorical features.

To train a Logistic Regression classification model.

To train a Decision Tree classification model.

To evaluate the models using accuracy, precision, recall, and F1-score.

To analyze model predictions using confusion matrices.

To develop an interactive Streamlit application for survival prediction.

2. Methodology

The Titanic Survival Prediction System follows a sequence of machine learning steps to transform the raw passenger dataset into a trained classification system.

Step 1: Input Dataset — The system uses the Titanic training dataset containing 891 passenger records and 12 columns. The Survived column is used as the target variable.

Step 2: Data Inspection — The dataset is inspected for column types, missing values, class distribution, and descriptive statistics.

Step 3: Feature Selection — The initial model uses Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked. PassengerId, Name, Ticket, and Cabin are excluded from the initial feature set.

Step 4: Missing Value Handling — Age is handled using median imputation and Embarked using most-frequent imputation inside the preprocessing pipeline. Cabin is excluded because 687 values are missing.

Step 5: Categorical Encoding — Sex and Embarked are converted into numerical representations using one-hot encoding.

Step 6: Numerical Scaling — Numerical variables are standardized for Logistic Regression using StandardScaler.

Step 7: Train-Test Split — The data is divided into 80% training data and 20% testing data using random_state=42 and stratification, producing 712 training and 179 testing records.

Step 8: Model Training — Two classifiers are trained: Logistic Regression and Decision Tree. The Decision Tree uses a maximum depth of 5.

Step 9: Model Evaluation — The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Step 10: Prediction Application — The trained Logistic Regression pipeline is integrated into a Streamlit application for interactive prediction.

Figure 1: Survival distribution in the Titanic dataset

3. Implementation

The system is implemented in Python using Pandas and Scikit-learn for data processing and machine learning. Matplotlib and Seaborn are used for visualization, and Streamlit is used to build the interactive application.

3.1 Technologies Used

3.2 Implementation Code

```
# Load dataset
df = pd.read_csv("train.csv")

features = [
    "Pclass", "Sex", "Age",
    "SibSp", "Parch", "Fare", "Embarked"
]

X = df[features]
y = df["Survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2,
    random_state=42, stratify=y
```

)

3.3 Working of the Program

The implementation begins by loading the Titanic training dataset. The selected columns are divided into input features X and target variable y. The data is split into training and testing subsets.

A ColumnTransformer separates numerical and categorical variables. Numerical variables are imputed using the median and standardized, while categorical variables are imputed using the most frequent category and one-hot encoded.

The transformed data is passed to the selected classifier. During prediction, the same preprocessing steps are automatically applied to new passenger data before the classifier produces a prediction.

3.2 Implementation Code

```
numeric_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer([
    ("num", numeric_pipeline, numeric_features),
    ("cat", categorical_pipeline, categorical_features)
])

model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression())
])

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

3.3 Working of the Program

The exploratory analysis showed an observed survival rate of approximately 74.2% for females and 18.9% for males. By passenger class, the observed rates were approximately 63.0% for first class, 47.3% for second class, and 24.2% for third class. These statistics describe associations in the dataset and do not establish causal effects.

Figure 2: Observed survival rate by gender
Figure 3: Observed survival rate by passenger class

3.4 Sample Input

A sample passenger can be entered into the Streamlit application using values such as:

3.5 Sample Output

The application returns a prediction in the form of Survived or Did Not Survive and displays the model's estimated probability of survival.

Example: Prediction: Survived — Probability of survival: model-generated percentage.

3.6 Advantages

- Simple and easy to implement.
- Uses standard machine learning preprocessing.
- Provides two interpretable classification approaches.
- Uses multiple evaluation metrics.
- Includes an interactive web application.
- Can be reproduced using the supplied dataset and code.

3.7 Limitations

The dataset is relatively small.

The initial model does not use all information contained in Name, Ticket, or Cabin.

Performance depends on the selected train-test split.

The model captures statistical patterns rather than causal relationships.

3.5 Sample Output

Model evaluation on the 179-record test set produced the following results.

Logistic Regression Confusion Matrix

Figure 4: Logistic Regression confusion matrix — [[98, 12], [23, 46]]

The Logistic Regression model produced 98 true negatives, 12 false positives, 23 false negatives, and 46 true positives. The model correctly classified 144 of 179 test passengers.

Decision Tree Confusion Matrix

Figure 5: Decision Tree confusion matrix — [[97, 13], [29, 40]]

The Decision Tree produced 97 true negatives, 13 false positives, 29 false negatives, and 40 true positives. The model correctly classified 137 of 179 test passengers.

4. Conclusion

The Titanic Survival Prediction System provides a practical demonstration of supervised machine learning for binary classification. The project covers data inspection, missing-value handling, categorical encoding, numerical scaling, exploratory analysis, model training, and evaluation.

Two models were implemented: Logistic Regression and Decision Tree. On the selected 80:20 test split, Logistic Regression achieved 80.45% accuracy and the Decision Tree achieved 76.54% accuracy. The evaluation also included precision, recall, F1-score, and confusion matrices.

The developed Streamlit application makes the trained model accessible through a simple user interface. A user can enter passenger characteristics and receive a model-generated survival prediction and estimated probability.

The project therefore demonstrates the complete workflow of a small machine learning application, from raw data to an interactive prediction system.

5. Future Scope

Feature engineering using passenger titles, family size, and group information.

Careful extraction of useful information from ticket and cabin data.

Cross-validation for more robust performance estimates.

Hyperparameter tuning of the Decision Tree and Logistic Regression models.

Experimentation with additional machine learning algorithms.

Enhanced Streamlit interface with interactive charts and improved input validation.

Model persistence and deployment improvements for production-style use.

Project Tables

Report Type	Project Report
Domain	Machine Learning / Data Science
Task	Binary Classification
Dataset	Titanic Passenger Dataset
Models	Logistic Regression, Decision Tree
Tools	Python, Pandas, Scikit-learn, Streamlit

Technology	Purpose
Python	Programming language and implementation
Pandas	Data loading and manipulation
Scikit-learn	Preprocessing, model training and evaluation
Logistic Regression	Binary classification model
Decision Tree	Rule-based classification model
Matplotlib / Seaborn	Data visualization and EDA
Streamlit	Interactive web application

Input	Example Value
Passenger Class	1
Sex	female
Age	25
Siblings/Spouses	0
Parents/Children	0
Fare	80
Embarked	C

Metric	Logistic Regression	Decision Tree
Accuracy	80.45%	76.54%
Precision – Class 0	0.81	0.77
Recall – Class 0	0.89	0.88
F1-score – Class 0	0.85	0.82
Precision – Class 1	0.79	0.75
Recall – Class 1	0.67	0.58
F1-score – Class 1	0.72	0.66

Figures





